STAFF REVIEW · H446 084 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Known best route cost from source.

2. g(n)+h(n).

3. Never overestimates true remaining cheapest cost.

4. mid3=8 then mid5=12; return index5.

I Do — source worked steps

1. Calculate all frontier f-scores — From P: Q has g=3,h=2,f=5 ; R has g=8,h=0,f=8 . Put both in OPEN. Choose Q next (smallest f), not simply the nearest-looking node.

2. Improve the path to R — Via Q→R, new g(R)=3+2=5 , which improves the existing 8. Thus f(R)=5+0=5 , and predecessor R becomes Q. OPEN now contains R with the improved values.

3. Confirm the path and heuristic meaning — Select goal R from OPEN, then reconstruct R←Q←P to obtain P→Q→R , cost 5 . The supplied heuristics do not overestimate remaining true cost; the next lesson can explore what happens when this fails.

We Do — guided CHECK

1. Compute priorities and choose
Calculate the two g/f pairs from the edges leaving X. State which candidate A* should expand next and why.
Expand Y
Y has g=4,h=3,f=7 , Z has g=10,h=0,f=10 , so expand Y. Via Y→Z, candidate g is 4+3=7: improve Z from 10 to 7, with predecessor Z←Y.
Finish the trace
Now Z has g=7,h=0,f=7 . Select it as the goal; predecessor chain Z←Y←X gives X→Y→Z , cost 7 .
Distinguish the actual path cost g from estimated priority f.

You Do — source mark schemes

1. B g2 [1],h2 [1],f4 [1]; C g5 [1],h1 [1],f6 [1].

2. C from5 to3 via B [2], D from6 to4 via C [2], A-B-C-D total4 [2].

3. With h0, f=g [2], same priority basis as Dijkstra [2], overestimating h can prioritise unsuitable paths and undermine optimality [2].

Exit Ticket — source answer

1. False. It is generally an estimate.